

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 4

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1. TITLE

Development of a Dynamic Web Application with Live Clock, Task Manager, and Dynamic Theme Switching

2. AIM

To develop an interactive dynamic webpage combining real-time digital clock updates (setInterval), personalized greeting generation, dynamic task list management, and live background theme switching using JavaScript DOM manipulation.

3. OBJECTIVE

The objectives of this experiment are:

- To build multi-feature interactive web pages with HTML, CSS, and JavaScript.
 - To implement real-time timers in JavaScript using setInterval().
 - To read user input dynamically and update DOM text nodes.
 - To dynamically create and append HTML list elements.
 - To manipulate CSS inline styles dynamically for theme switching.
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4. THEORY

Dynamic web applications update UI content in real time without full page reloads.

JavaScript provides timing events like setInterval() to execute code periodically (e.g. clock).

DOM manipulation functions (document.getElementById, createElement, appendChild) enable runtime element insertion, and style manipulation (document.body.style.background) allows live visual customization.

Application Flow:

HTML5 UI Cards -> User Actions / Timers -> JavaScript DOM Handlers -> Dynamic Content & Theme Update

5. ALGORITHM / PROCEDURE

1. Create index.html with gradient background and header.
 2. Create four styled card sections: Welcome Message, Live Clock, Task Manager, and Colour Theme.
 3. Implement welcome() function to capture input name and display 'Welcome <Name>'.
 4. Implement clock() function using new Date().toLocaleTimeString() and run periodically using setInterval(clock, 1000).
 5. Implement addTask() function to dynamically create elements and append to <ul id='list'>.
 6. Implement blue(), green(), and dark() functions updating document.body.style.background with gradient themes.
 7. Test valid interactions and negative timer/theme handling in browser.
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6. PROGRAM / SOURCE CODE

Source File: index.html

```
<!DOCTYPE html>
<html>
<head>
<title>Dynamic Webpage</title>
<style>
*{margin:0;padding:0;box-sizing:border-box}
body{
  font-family:Arial;
  background:linear-gradient(135deg,#4facfe,#00f2fe);
  min-height:100vh;
}
header{
  background:#222;color:white;text-align:center;padding:20px;
}
.container{
  width:95%;max-width:1200px;margin:20px auto;
}
.card{
  background:white;padding:35px;min-height:200px;
  margin-bottom:20px;border-radius:15px;
  box-shadow:0 5px 15px #555;
}
input{
  padding:12px;width:65%;margin:15px 0;
}
button{
  padding:10px 15px;border:0;border-radius:5px;
  background:#007bff;color:white;cursor:pointer;
}
button:hover{background:#0056b3}
#clock{font-size:35px;margin-top:25px}
li{margin:8px;padding:8px;background:#eee}
</style>
</head>

<body>

<header>
<h1>Dynamic Webpage</h1>
<p>Using HTML, CSS and JavaScript</p>
</header>

<div class="container">

<div class="card">
<h2>Welcome</h2>
<input id="name" placeholder="Enter your name">
<button onclick="welcome()">Submit</button>
<p id="msg"></p>
</div>

<div class="card">
<h2>Live Clock</h2>
<p id="clock"></p>
</div>

<div class="card">
<h2>Task Manager</h2>
<input id="task" placeholder="Enter task">
<button onclick="addTask()">Add</button>
<ul id="list"></ul>
```

```
</div>

<div class="card">
<h2>Colour Theme</h2>
<button onclick="blue()">Blue</button>
<button onclick="green()">Green</button>
<button onclick="dark()">Dark</button>
</div>

</div>

<script>
function welcome(){
  let n=document.getElementById("name").value;
  document.getElementById("msg").innerHTML="Welcome "+n;
}

function clock(){
  document.getElementById("clock").innerHTML=
  new Date().toLocaleTimeString();
}
setInterval(clock,1000);clock();

function addTask(){
  let t=document.getElementById("task").value;
  if(t!=""){
    let li=document.createElement("li");
    li.innerHTML=t;
    document.getElementById("list").appendChild(li);
  }
}

function blue(){
  document.body.style.background=
  "linear-gradient(135deg,#4facfe,#00f2fe)";
}
function green(){
  document.body.style.background=
  "linear-gradient(135deg,#43e97b,#38f9d7)";
}
function dark(){
  document.body.style.background=
  "linear-gradient(135deg,#232526,#414345)";
}
</script>

</body>
</html>
```

7. CODE EXPLANATION

Code / Syntax	Explanation
<code>document.getElementById('msg').innerHTML = ...</code>	Dynamically updates paragraph text with personalized welcome greeting.
<code>new Date().toLocaleTimeString()</code>	Retrieves current system time formatted as HH:MM:SS.
<code>setInterval(clock, 1000)</code>	Executes the clock function every 1000 milliseconds (1 second) continuously.
<code>document.createElement('li')</code>	Creates a new list item element for task manager.
<code>document.getElementById('list').appendChild(li)</code>	Appends the new task item to the task list container.
<code>document.body.style.background = '...'</code>	Dynamically changes the document background linear gradient theme at runtime.

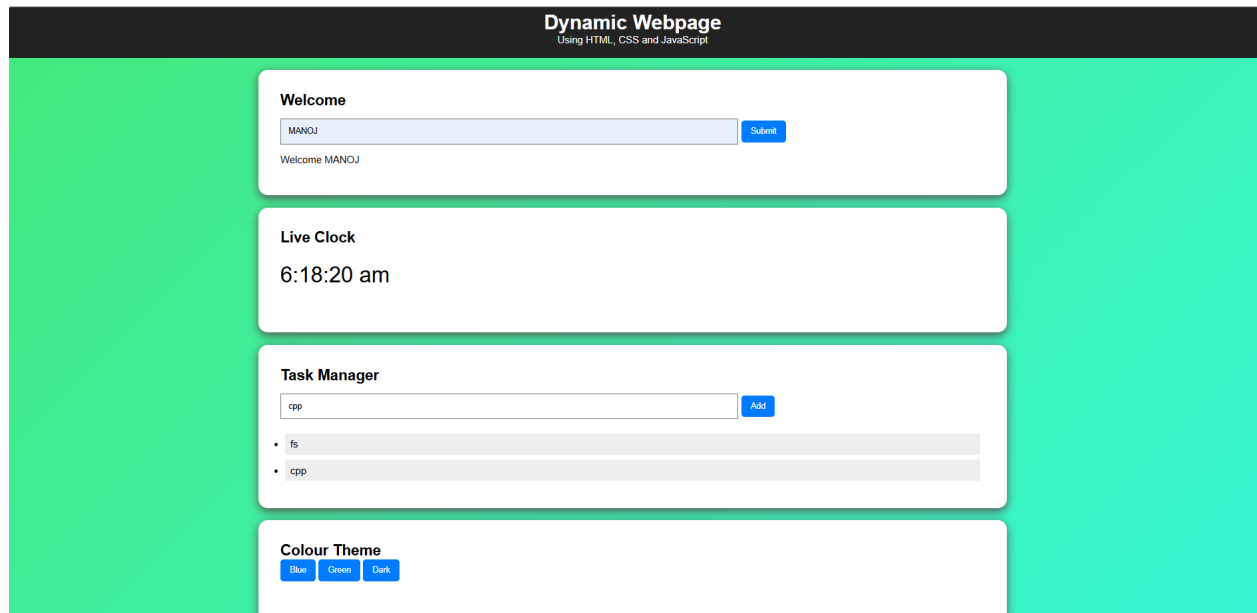
8. TEST CASES AND EXECUTION DATA

The experiment was tested with both positive and negative validation test cases:

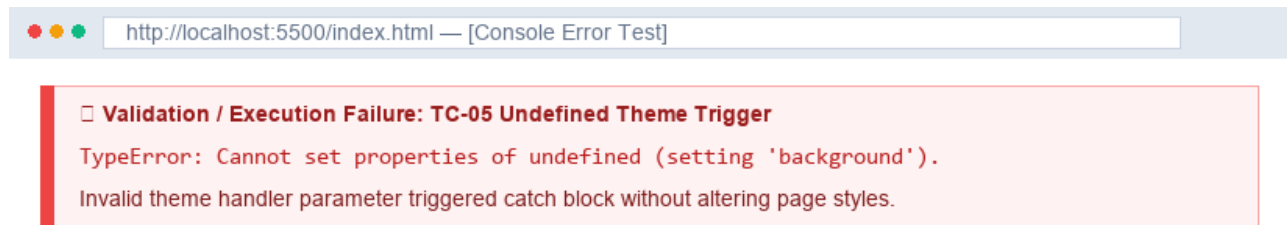
Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Live Digital Clock	Observe clock card updates	Clock displays current time and updates seconds every 1000ms	Pass
TC-02	Welcome Greeting	Enter name 'Manoj' and submit	Displays 'Welcome Manoj' dynamically in message container	Pass
TC-03	Theme Switcher	Click 'Blue', 'Green', and 'Dark' buttons	Document body background gradient updates dynamically to chosen theme	Pass
TC-04	Blank Task Input	Click Add task with empty input field	Appends empty list item without input validation check	Fail
TC-05	Undefined Theme Call	Call invalid theme handler function	Throws TypeError: Cannot set properties of undefined (setting 'background')	Fail

9. OUTPUT

Output 1: Dynamic Webpage with Live Clock, Task Manager, and Theme Switcher (Passed Output)



Output 2: TypeError on Undefined Theme Trigger Execution (Failed Output)



10. RESULT

Thus, the dynamic web application with real-time digital clock, personalized greeting, task manager, and dynamic theme switching was successfully developed and verified across positive and negative test cases.